



UTAH

Multi-agent Reinforcement Learning (MARL) for Open Agent Systems

MOASEI AAMAS-2025 Competition

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Introduction

- **Open Agent Systems (OASYS):**
 - Models real-world non-stationarity
 - Systems that allow such dynamic participation

Track #3: *Wildfire*

- **Agent openness:** available agents appear/disappear
- **Task openness:** available tasks appear/disappear

Related Work: Graph Neural Networks (GNNs)

- **Graph Neural Networks (GNNs)**
 - Deep learning models for graph-structured data
 - Leverage node features, edge connections, and global graph attributes
- **Message Passing Framework**
 1. Message Computation: Aggregate feature messages from each neighbor
 2. Aggregation: Sum or average the incoming messages
 3. Update: Combine the aggregated message with the node's own features to produce a new embedding
- **Layer Stacking**
 - Stack multiple message-passing layers to capture multi-hop relationships
- **Common Variants**
 - GCN (Graph Convolutional Network): Simple sum/average aggregation
 - GAT (Graph Attention Network): Learns attention weights over neighbors
 - GraphSAGE: Sampling-based neighborhood aggregation with flexible functions

Related Work: Graph Attention Network (GAT)

- Learns **attention weights** to prioritize important neighbors

1. Attention Mechanism

- Feature transformation:**

$$\mathbf{h}'_i = W \mathbf{h}_i$$

- Unnormalized attention score:**

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{h}'_i \parallel \mathbf{h}'_j])$$

2. Normalized attention coefficient:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}$$

3. Node Update

$$\mathbf{h}_i^{\text{new}} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{h}'_j \right)$$

where σ is an activation (e.g., ReLU).

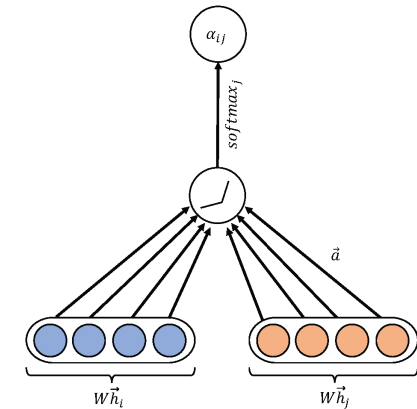
4. Multi-Head Attention

- Learn K independent attention heads
- Concatenate:**

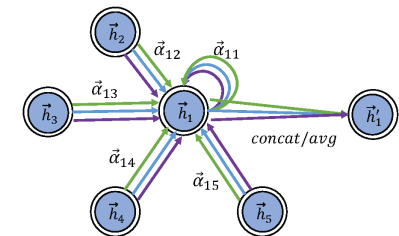
$$\parallel_{k=1}^K \mathbf{h}_i^{(k)}$$

- Average:**

$$\frac{1}{K} \sum_{k=1}^K \mathbf{h}_i^{(k)}$$



(a) Attention mechanism



(b) Multihead attention

Figure 1: The illustration of the GAT model

Problem Statement

- S : State
(agent positions, suppressant levels; fire locations, intensities)
- U : Actions (suppress fire, no-op)
- P : State transition function
- r : Shared team reward
- Z : Observation space
- O : Observation function
- n : the number of active agents

$$M = \langle S, U, P, r, Z, O, n \rangle$$

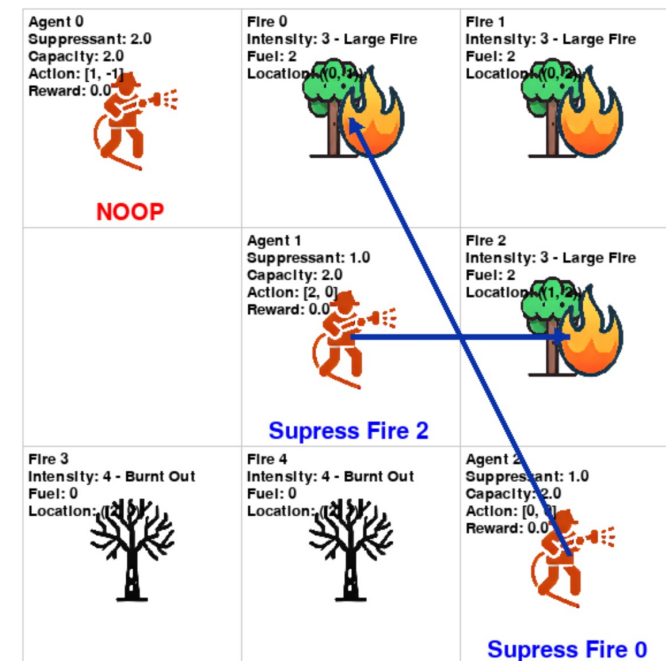


Figure 2: Wildfire Environment

Method

- GNN-based Advantage Actor-Critic (A2C)
- GNN-based policy learning with Multi-objective Heads
 - GNN with Vanilla Policy Gradient (VPG)
 - GNN with Suppressant level Prediction (SP)
 - GNN with Intent Prediction and Belief Update (IP + BU)

Method: State Representation

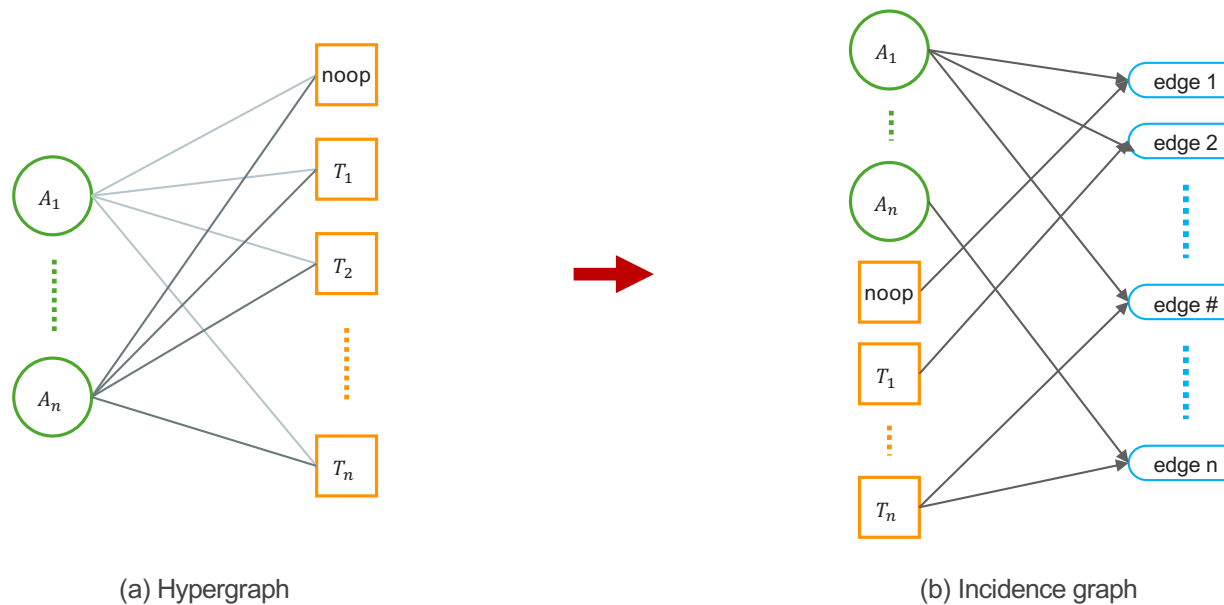


Figure 3: Incidence graph of the hypergraph

Method: Advantage Actor-Critic (A2C)

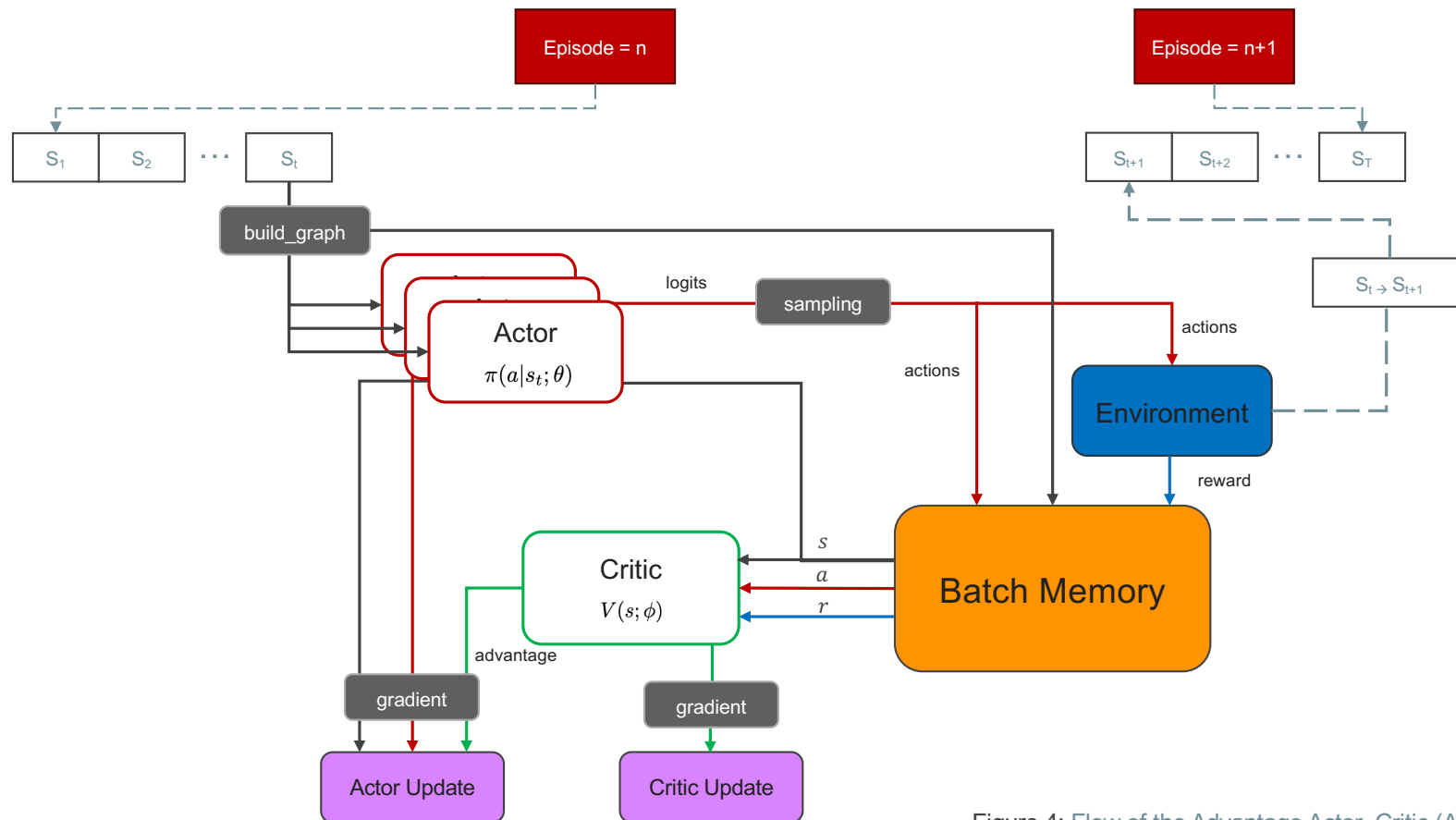


Figure 4: Flow of the Advantage Actor-Critic (A2C)

Method: 1) GNN-based A2C

- Discounted returns

$$R_t = \sum_{k=0}^{T-t} \gamma^k r_{t+k}$$

- Advantage

$$A_t = R_t - V(s_t)$$

- Loss

$$\mathcal{L}_{\text{actor}} = -\log \pi(a_t | s_t) A_t$$

$$\mathcal{L}_{\text{critic}} = (V(s_t) - R_t)^2$$



Figure 5: Advantage Actor-Critic (A2C) Model

Method: 2) GNN-based policy learning

- $\mathcal{L}_{\text{task}}$: Task selection (RL loss)
- $\mathcal{L}_{\text{suppress}}$: Predict suppressant level of other agents
- $\mathcal{L}_{\text{intent}}$: Predict intent action of other agents
- $\mathcal{L}_{\text{belief}}$: Penalty for overconfident fire predictions

$$\mathcal{L}_{\text{task}} = -\log \pi(a_i | o_i) \cdot R_i$$

$$\mathcal{L}_{\text{suppress}} = \sum \text{CrossEntropy}(s_{\text{true}}, s_{\text{pred}})$$

$$\mathcal{L}_{\text{intent}} = \sum \text{CrossEntropy}(\hat{a}_j, a_j)$$

$$\mathcal{L}_{\text{belief}} = \frac{1}{|T|} \sum_{t \in T} (b_t \cdot (1 - o_t))^2$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{suppress}} \mathcal{L}_{\text{suppress}} + \lambda_{\text{intent}} \mathcal{L}_{\text{intent}} + \lambda_{\text{belief}} \mathcal{L}_{\text{belief}}$$

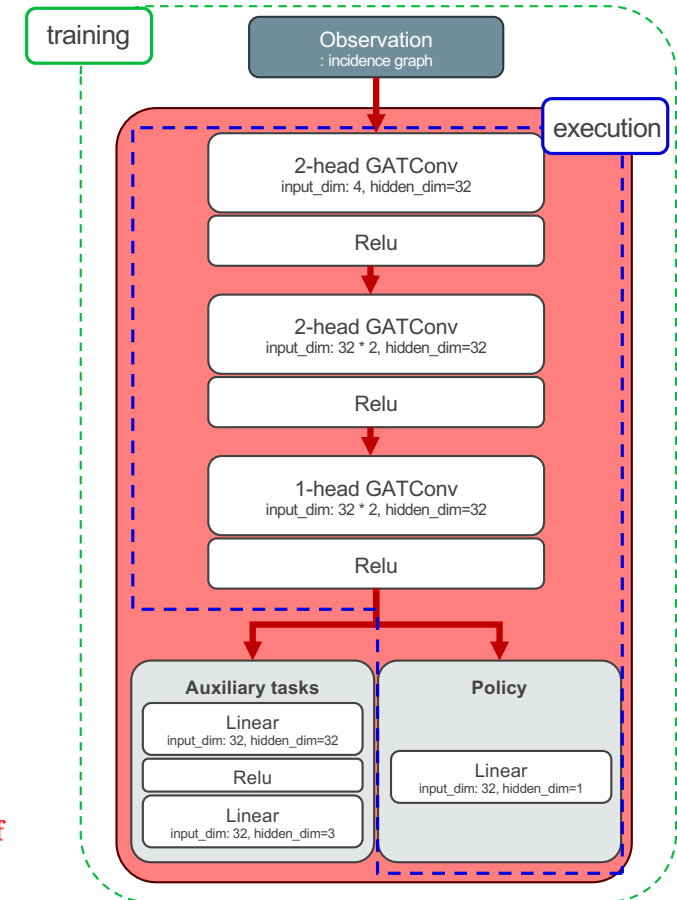
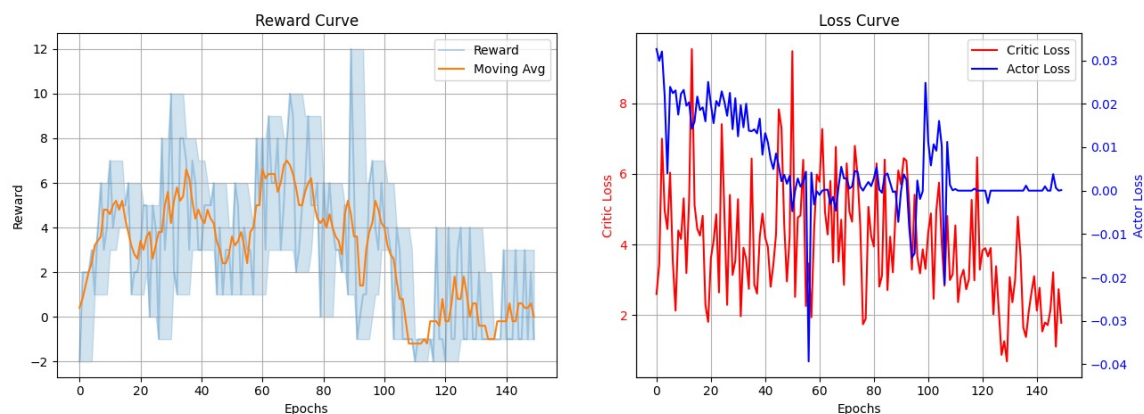


Figure 6: GNN-based policy learning Model

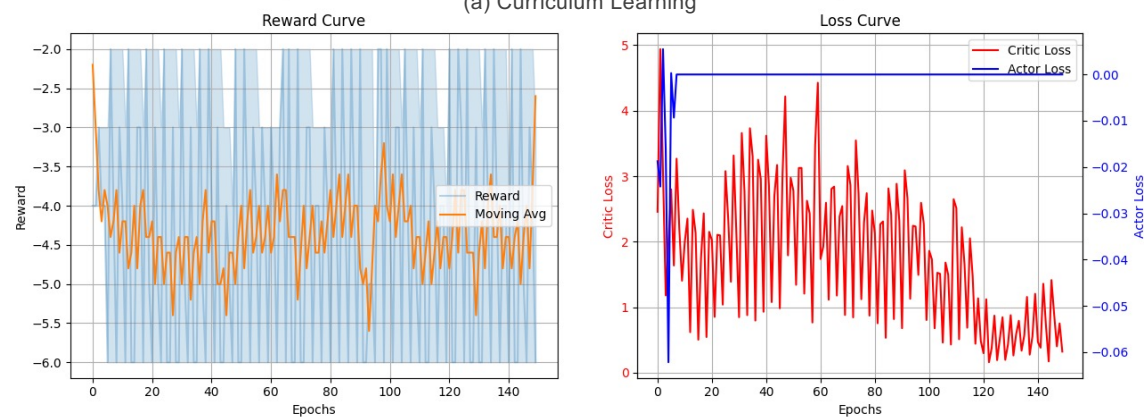
Experiment

- Curriculum Learning:
 - WS1 -> WS2 -> WS3
 - Random Sequence
- Algorithm Comparison
 - A2C
 - Policy learning
- Loss Analysis
 - Task loss
 - Suppressant loss
 - Intent loss
 - Belief loss
- By Configurations
 - Total reward
 - Average reward

Experiment: Curriculum Learning



(a) Curriculum Learning



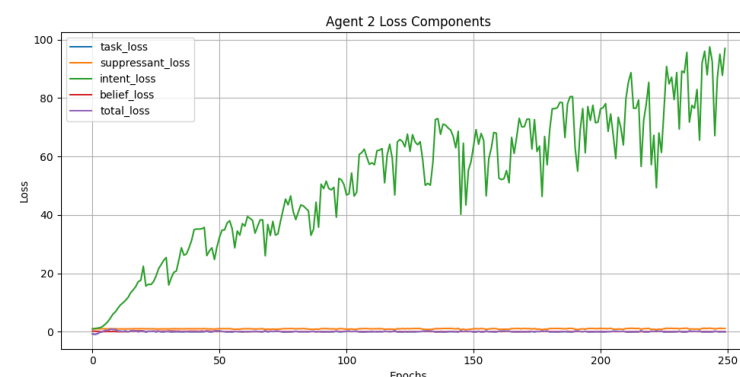
(b) Random Sequence

Figure 7: A2C training reward and loss curves

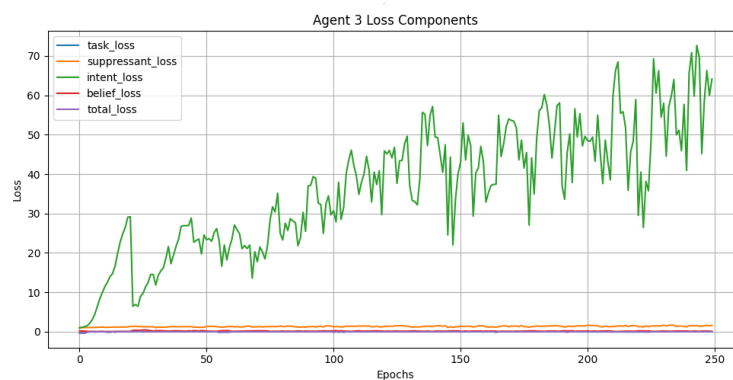
Experiment: Loss Analysis



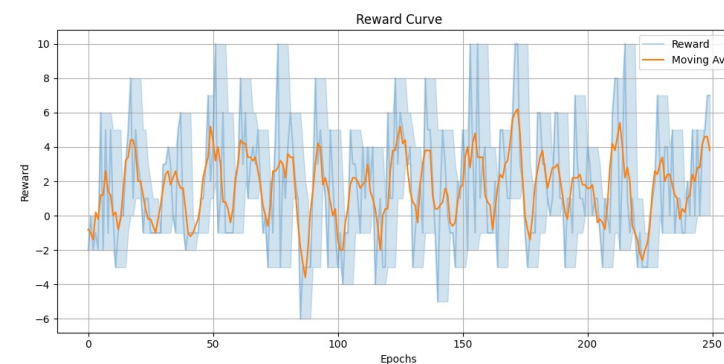
(a) firefighter1 Loss Components



(b) firefighter2 Loss Components



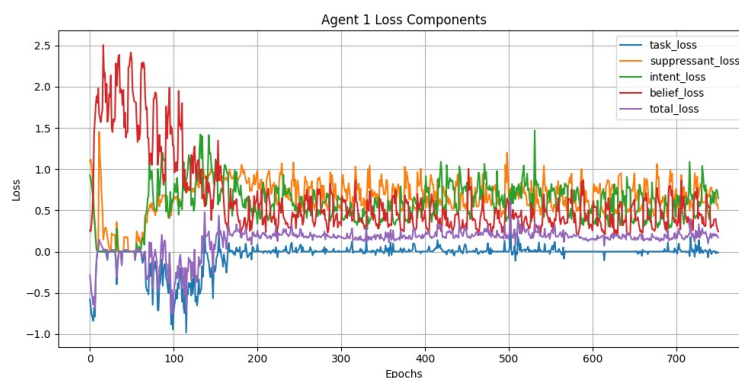
(c) firefighter3 Loss Components



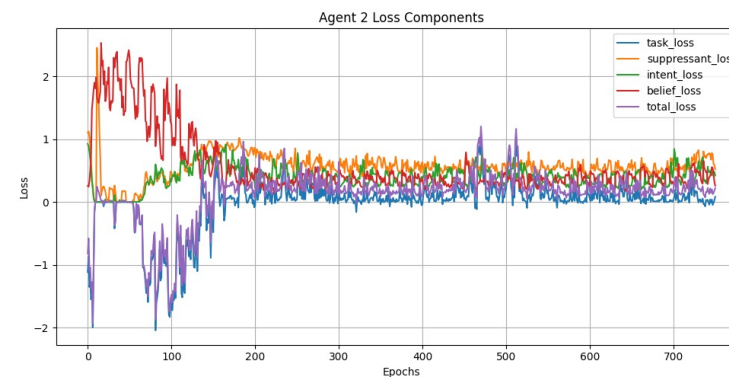
(d) Reward Curve

Figure 8: Individual loss curves for each firefighter agent under policy learning

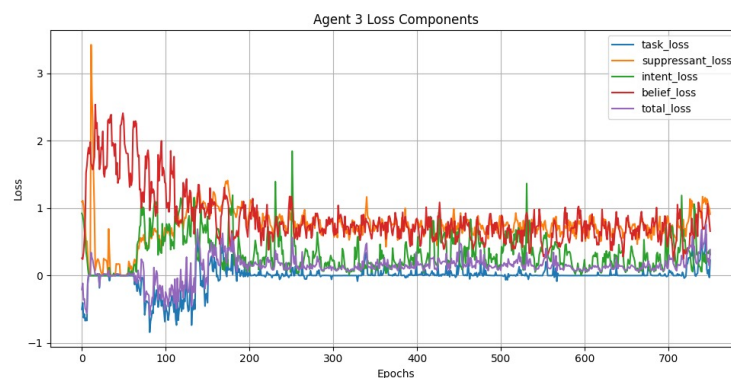
Experiment: Loss Analysis



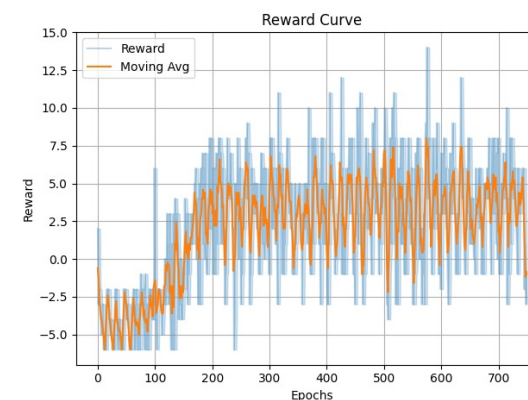
(a) firefighter1 Loss Components



(b) firefighter2 Loss Components



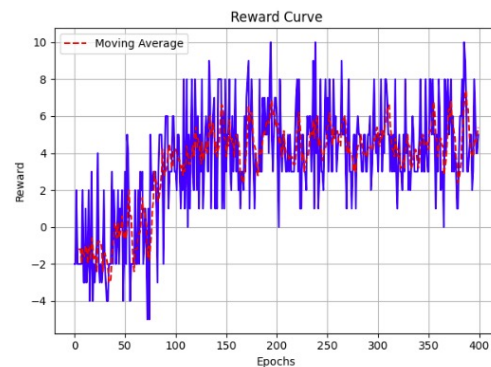
(c) firefighter3 Loss Components



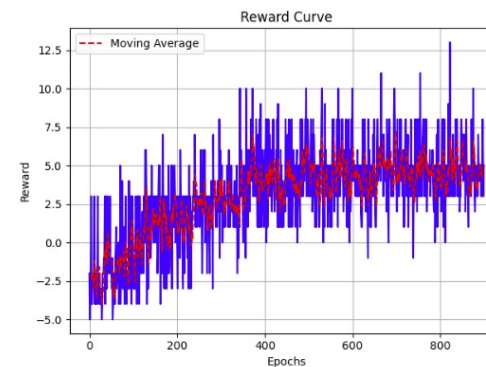
(d) Reward Curve

Figure 10: Individual loss curves for each firefighter agent under policy learning

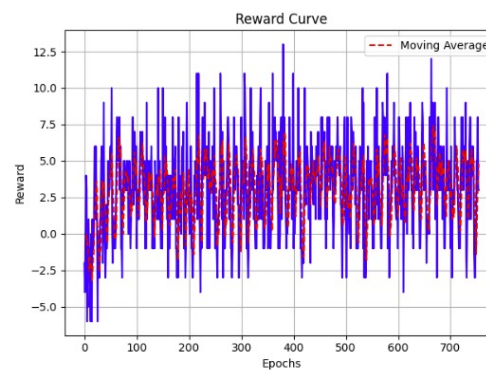
Experiment: Loss Analysis



(a) GNN-VPG



(b) + Suppressant Prediction



(c) + Intent + Belief Update

Figure 9: Policy learning reward curves during training

Experiment: Algorithm Comparison

Model	WS1		WS2		WS3	
	Avg Reward	Total Reward	Avg Reward	Total Reward	Avg Reward	Total Reward
GNN – A2C	4.7 ± 1.70	14.1 ± 5.11	4.1 ± 3.04	12.3 ± 9.11	0.0 ± 2.11	0.0 ± 6.33
GNN + VPG	4.4 ± 1.71	13.2 ± 5.14	5.1 ± 3.70	10.3 ± 11.09	-0.6 ± 1.65	1.8 ± 4.94
GNN + SP	5.0 ± 2.16	15.0 ± 6.48	4.9 ± 2.89	14.7 ± 8.65	0.8 ± 3.08	2.4 ± 9.25
GNN + SP + IP + Belief	4.7 ± 1.70	14.1 ± 5.11	4.7 ± 2.06	14.1 ± 6.17	0.8 ± 1.93	2.4 ± 5.80

Table 1: Comparison of Avg. and Total Rewards across wildfire scenarios WS1–WS3

Challenges and Insights

- Challenges

- Agents and tasks change dynamically, breaking standard MARL assumptions.
- Agents must coordinate without explicit communication or shared state.
- The environment is highly non-stationary, often destabilizing traditional learning methods.

- Insights

- GNNs capture dynamic agent-task relations effectively.
- Adding suppressant, intent, and belief heads improves coordination.
- A centralized training setup with decentralized execution generalizes across dynamic teams.

Conclusion

- Wildfire testbed for evaluating open-system MARL methods
- Decentralized agents learn adaptive, communication-free policies in dynamic settings
- Auxiliary objectives (suppressant, intent, belief) boost learning and coordination
- Higher training variance but stable, robust test performance

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Q&A

Thank you

Varun Raveendra, Yanxi Lin and Seongil Heo